NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

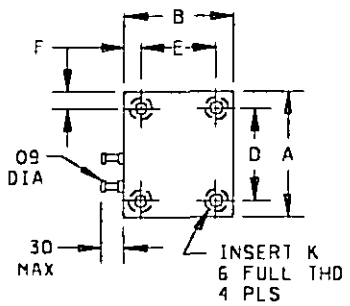
MIL-PRF-27/337D
22 January 1992
SUPERSEDING
MIL-T-27/337C
17 January 1986

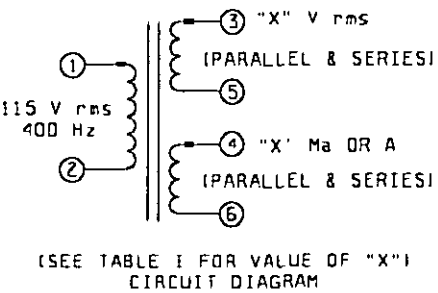
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 150 AND 200 VOLT AMPERES, 400 HZ, TF5S03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





(SEE TABLE I FOR VALUE OF "X")
CIRCUIT DIAGRAM

Inches	mm
.02	0.5
.09	2.3
.30	7.6
.40	10.2

Ltr	Dimensions	
	Inches	mm
A	2.43 ±.02	(50.8) ±(.5)
B	2.00 ±.02	(41.1) ±(.5)
C	2.10 ±.02	(46.0) ±(.5)
D	1.93 ±.02	(38.1) ±(.5)
E	1.50 ±.02	(28.4) ±(.5)
F	.25 ±.04	(6.4) ±(1.0)
H	1.90 ±.04	(38.1) ±(1.0)
J	.21 ±.04	(42.9) ±(1.0)
K	.138-32 UNC 2B	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the top and sides of case.
4. Dimensional tolerance shall be ±.02 unless otherwise specified.
5. For series output connect terminals 4 and 5 together. For parallel output connect pins 3 to 4 and 5 to 6.

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the winding)

Electrical ratings.

Primary voltage (1-2) 115 volts rms, 400 ±20 hertz

Secondary voltage and current (3-6) See table I

Voltampere See table I

Working voltage (1-2). 535 volts peak

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case. Encapsulated

Material Epoxy fiberglass

Terminals Turret type

Height 300 inch, maximum

Weight 2 pounds, maximum (PIN's -01 through -10) and 2 2 pounds, maximum (PIN's -11 through -19)

Operating temperature range -55°C to +130°C

Altitude 70,000 feet

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage (each winding, secondary windings connected in series)

At sea level 1500 volts rms

At reduced barometric pressure. 1 25 times the peak working voltage specified

Electrical characteristics

Rated load With 115 volts rms and 400 Hz in (1-2), and rated current in secondary, the voltage across (3-6) shall be as specified in table I, 5 percent.

Efficiency 90 percent minimum at rated load

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TABLE I Electrical ratings.

Dash number M27/337	Secondary voltage (3-6) (V rms) ±5%		Secondary current (3-6) (amperes)		Volt-ampere rating
	Series (CT)	Parallel	Series	Parallel	
-01	10	5	15	30	150
-02	12 6	6 3	11 9	23.8	150
-03	16	8	9 37	18 7	150
-04	20	10	7 50	15	150
-05	24	12	6.25	12 5	150
-06	34	17	4 41	8 82	150
-07	40	20	3 75	7 50	150
-08	56	28	2 68	5 36	150
-09	88	44	1 70	3 40	150
-10	120	60	1 25	2 50	150
-11	12.6	6 3	15 85	31 70	200
-12	16	8	12 50	25	200
-13	20	10	10	20	200
-14	24	12	8 35	16 70	200
-15	34	17	5 90	11 80	200
-16	40	20	5	10	200
-17	56	28	3.57	7 14	200
-18	88	44	2.27	4 54	200
-19	120	60	1 66	3 53	200

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Temperature rise 45°C with 115 volts rms, 400 hertz across (1-2) at an ambient temperature of 85°C, full load terminals (3-6), for PIN's -01 through -10.

Temperature rise 45°C with 115 volts rms, 400 hertz across (1-2) at an ambient temperature of 70°C, full load terminals (3-6), for PIN's -11 through -19

Regulation $\frac{\text{Voltage (no load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$ (Shall not exceed 10 percent)

Polarity Additive, with terminals 2 and 3, and 5 and 4 connected

Marking location See figure 1

Part or Identifying Number (PIN) M27/337-(dash number from table I)

QUALITY ASSURANCE PROVISIONS

Extent of qualification

Qualification testing and approval to M27/332-10 and M27/337-19 shall be sufficient to grant qualification approval to MIL-T-27/332 through MIL-T-27/337 inclusive, all parts

Qualification testing and approval to M27/337-19 shall be sufficient to grant qualification approval to M27/337-01 through M27/337-19

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0772)

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